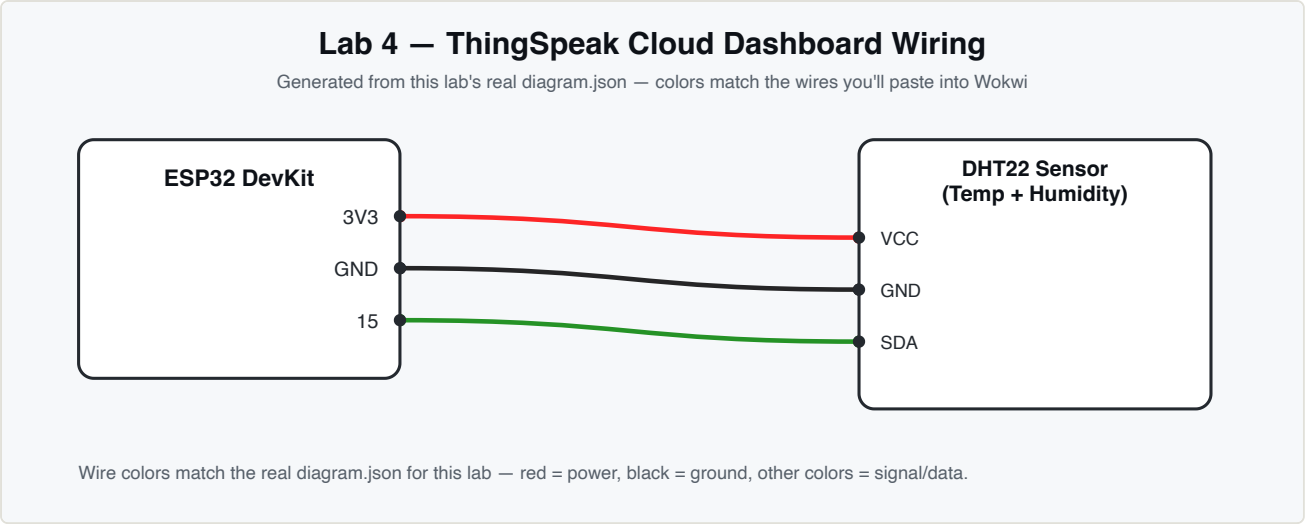
Lab 4 — Cloud Dashboard with ThingSpeak

Time: ~22 minutes in the Telecom & Data-Centre Edition (~30 minutes on your own) **You'll learn:** how to send sensor data to a cloud platform over HTTP, and get a live graph without writing any dashboard/website code yourself.

> **Telecom & Data-Centre Edition**
>
> **The story:** This is the trend view of one tower site's shelter, for the NOC and for management: temperature and humidity over time.
>
> **Before the day:** Create your free ThingSpeak account in advance (see the getting-started guide), so the lab time goes on the dashboard, not on sign-up emails.
>
> **Your team:** Teams of three. Rotate the roles from the last lab: **builder** (keyboard), **analyst** (reads the guide, checks the output), **decision owner** (answers the manager task).
>
> **Manager task:** Pick the 3 numbers you would show the board every month about the tower network, and say why.
>
> **Extension (fast teams):** On your channel's **Private View**, click **Add Widgets**, choose **Gauge**, and show Field 1 (temperature) with a red range above 28 °C. No code needed.
>
> **Business-case canvas:** Fill in box 5 (the KPI that proves it works).

Wiring diagram

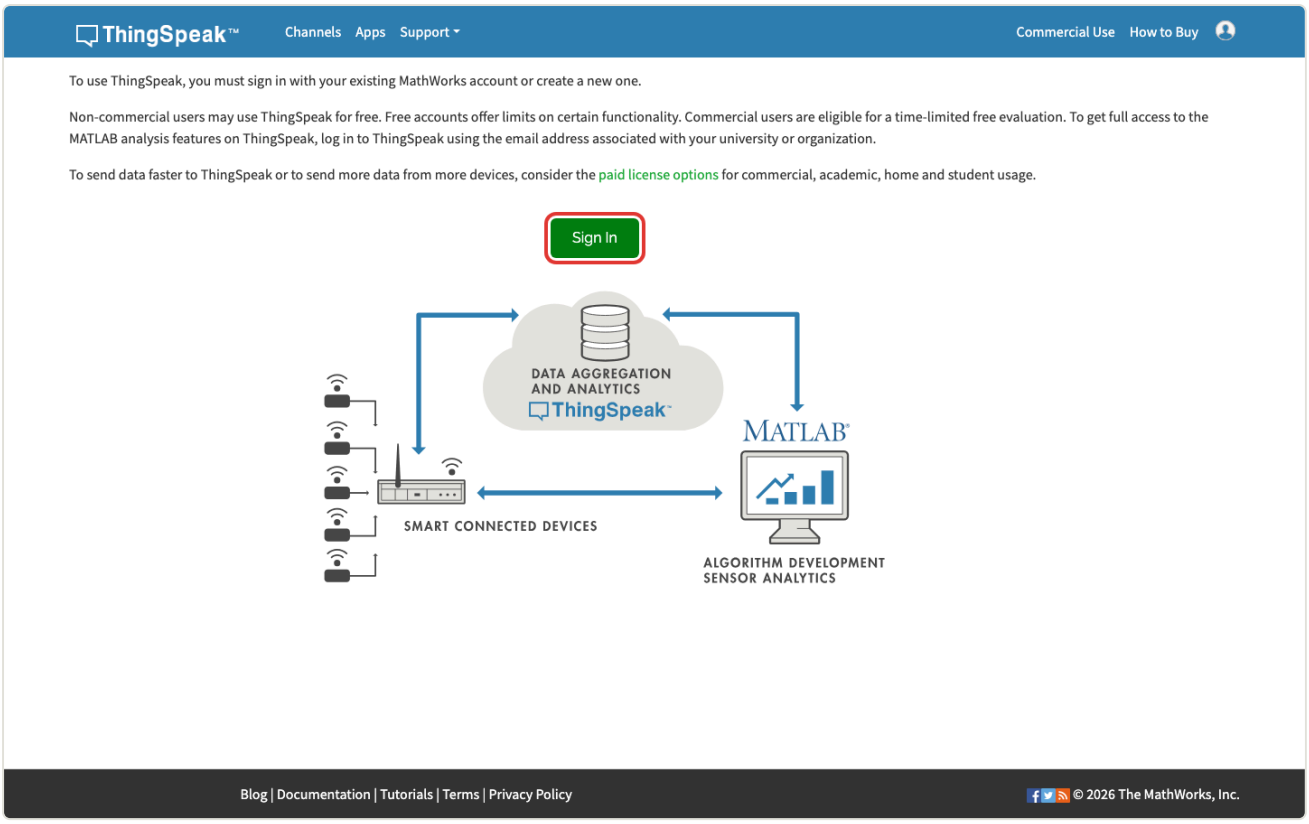


ESP32 Pin	Wire Color	Connects To
`3V3`	red	DHT22 Sensor (VCC)
`GND`	black	DHT22 Sensor (GND)
`15`	green	DHT22 Sensor (data pin)

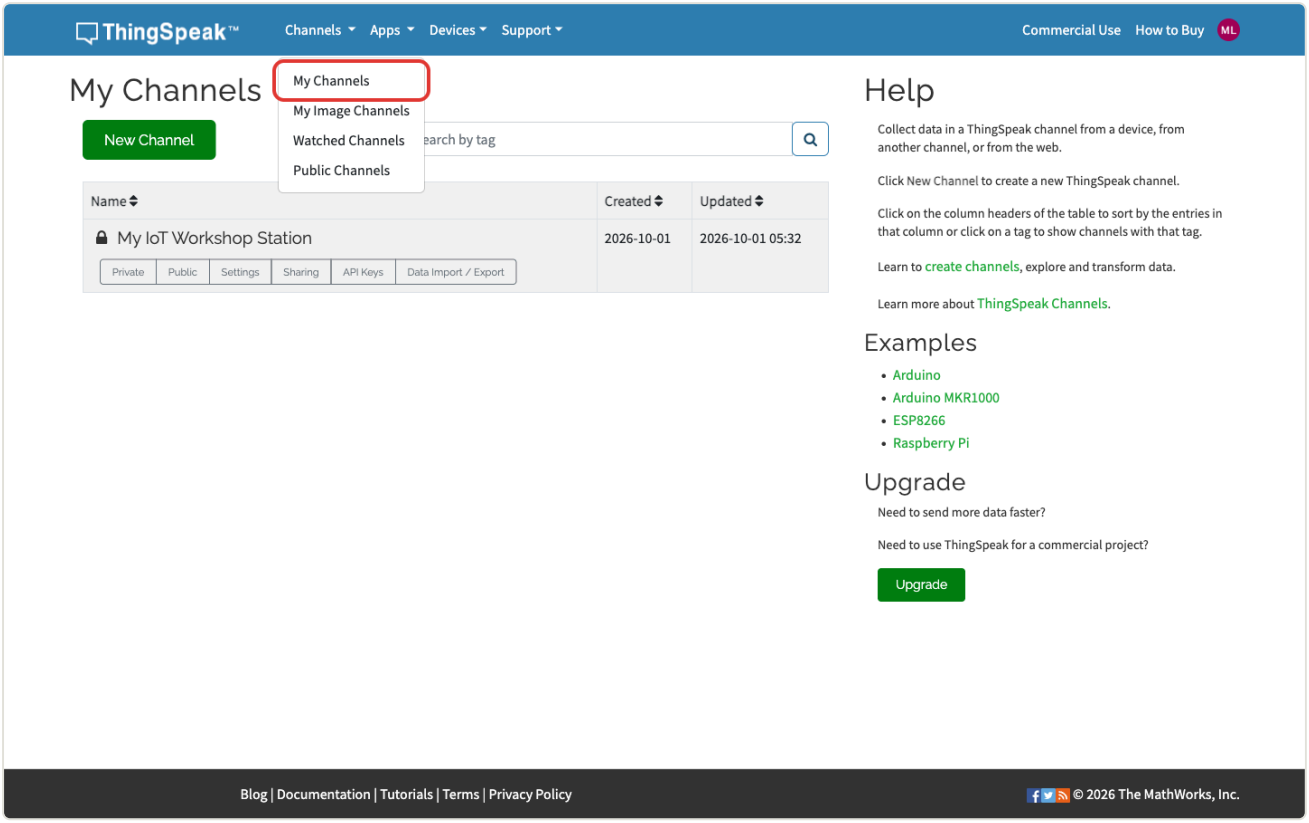
Same wiring as Labs 2 and 3 — this lab is entirely about what the code does with the reading once it has one.

Part A — Create your ThingSpeak channel (5 min)

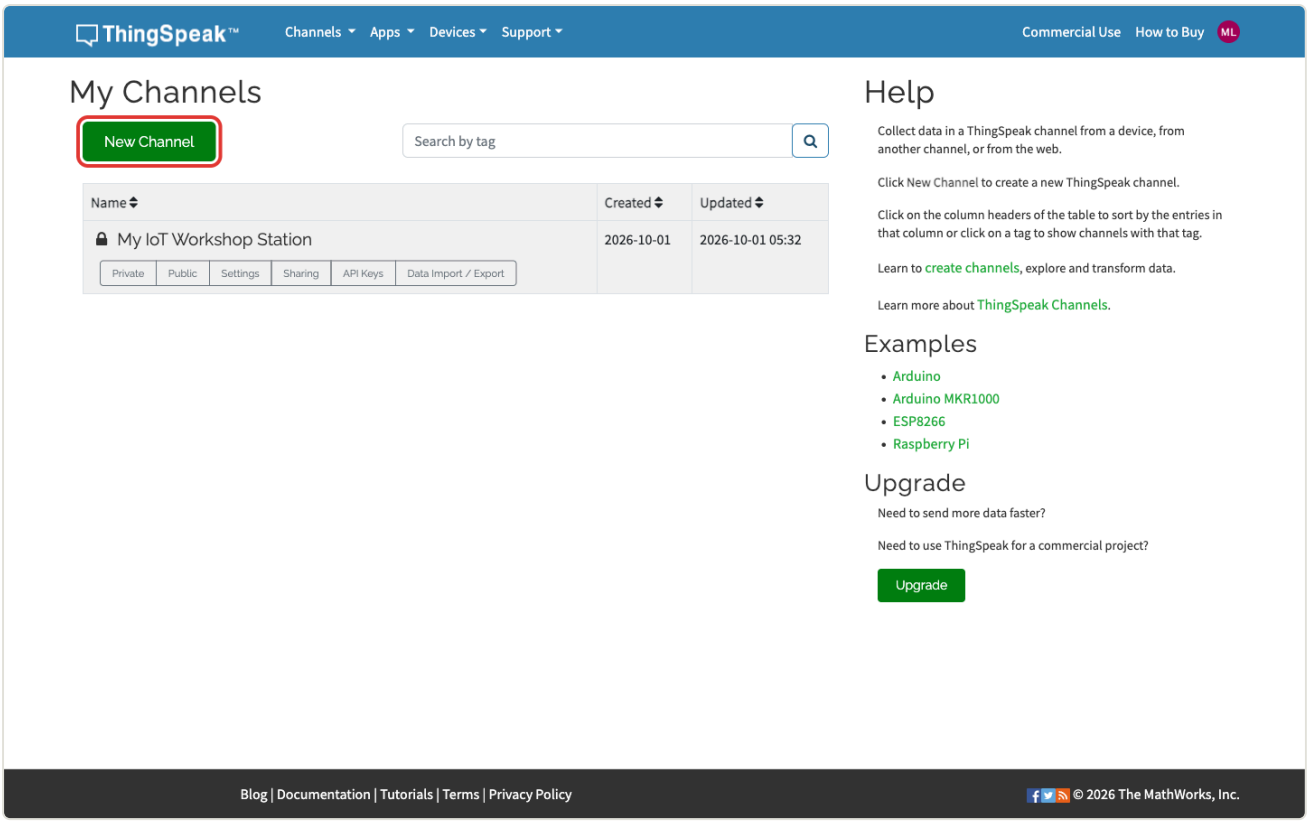
1. Go to thingspeak.com and click **Sign In**. Use the free MathWorks account you made in the Getting Started guide.



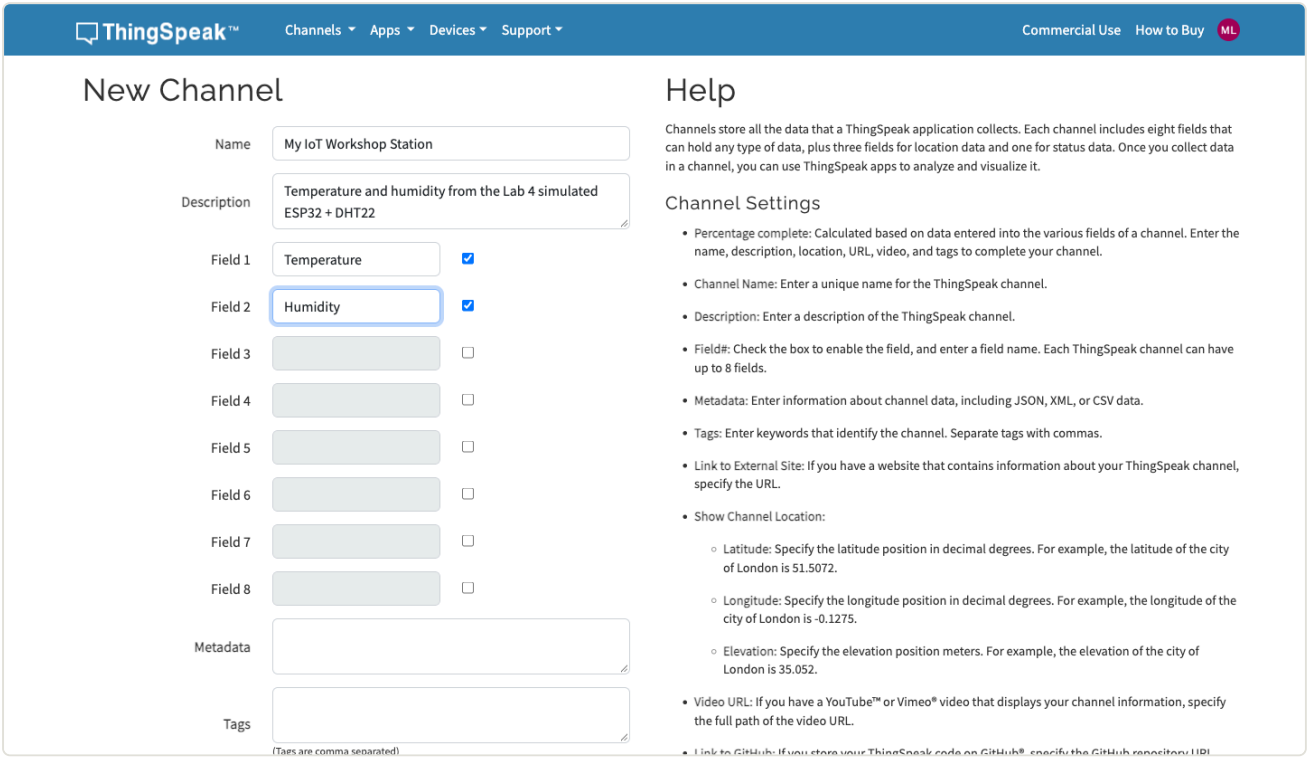
2. Click **Channels → My Channels**.



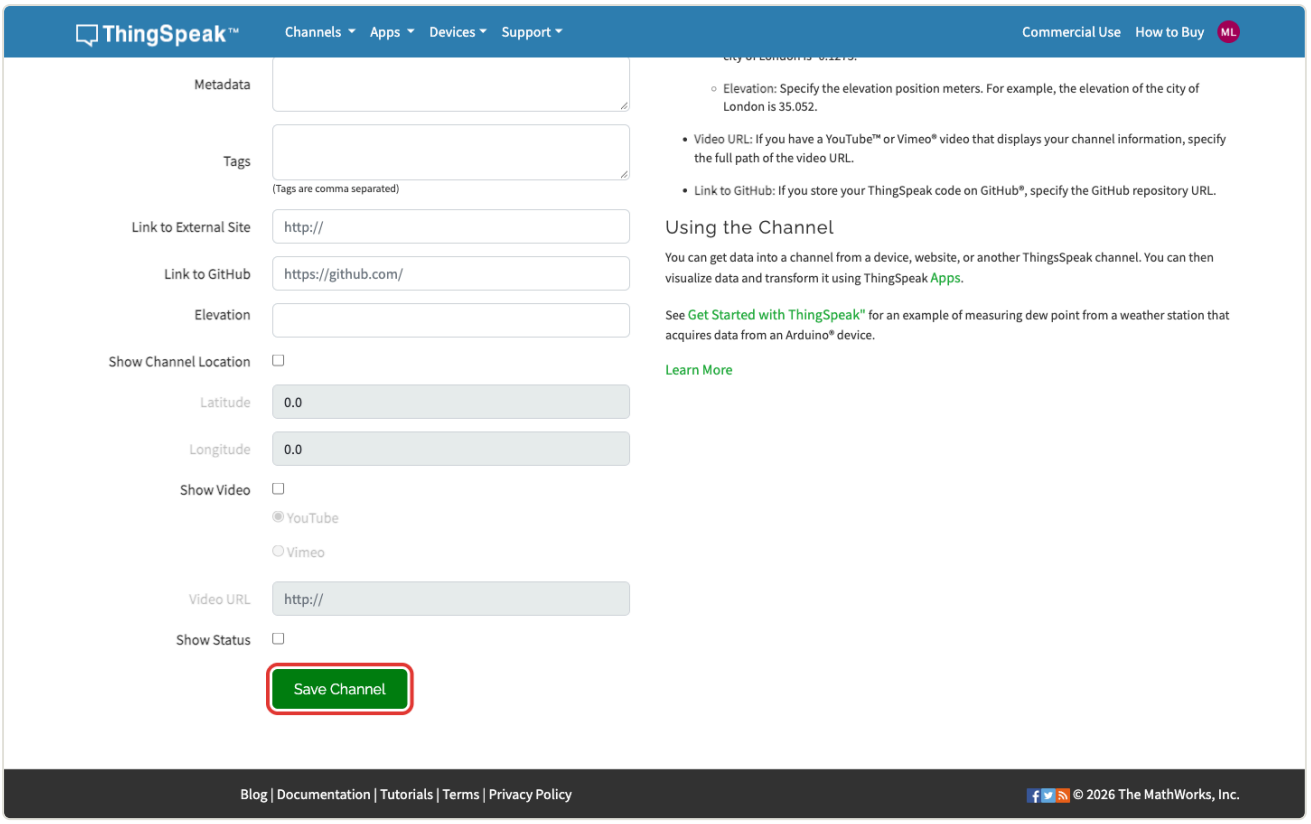
3. Click **New Channel**.



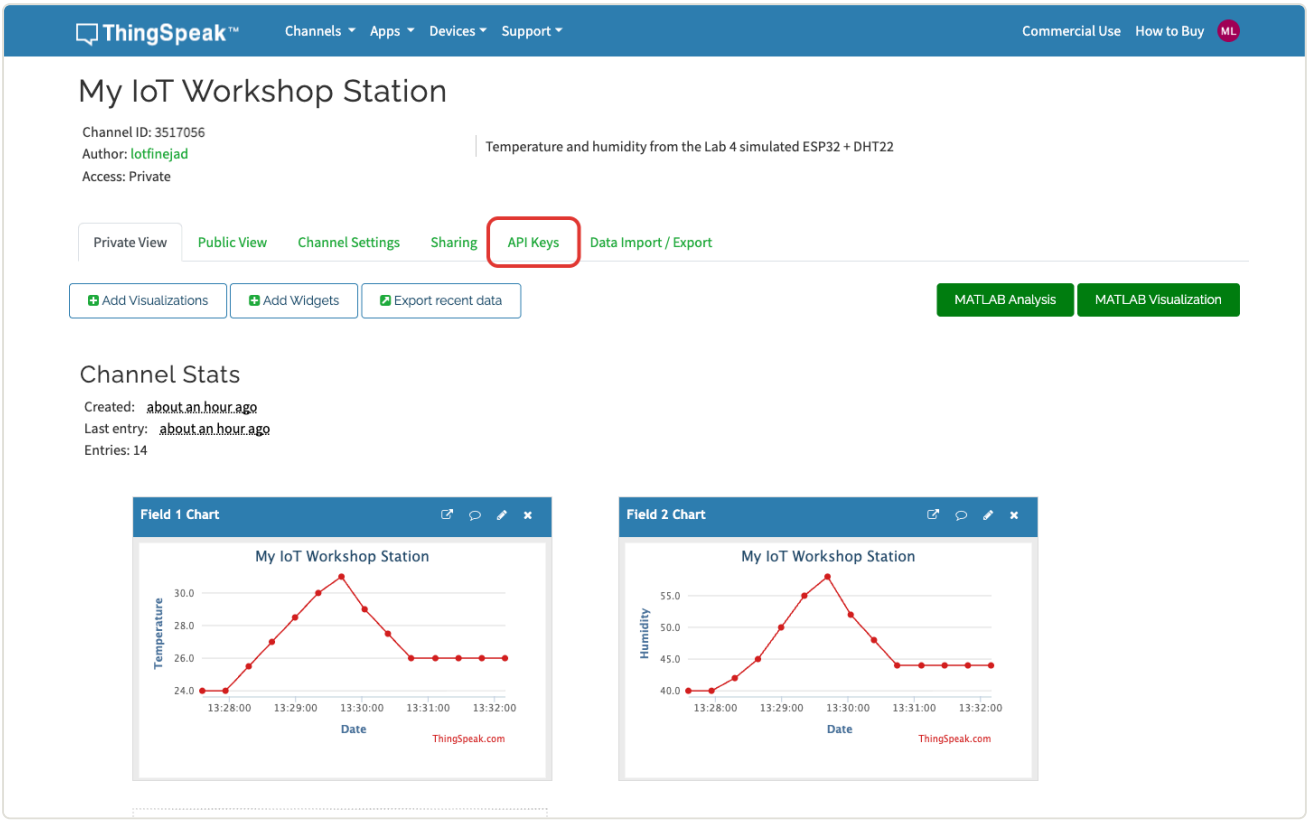
4. Fill in the form: - **Name:** for example “My IoT Workshop Station”. - Tick **Field 1** and name it `Temperature`. - Tick **Field 2** and name it `Humidity`.



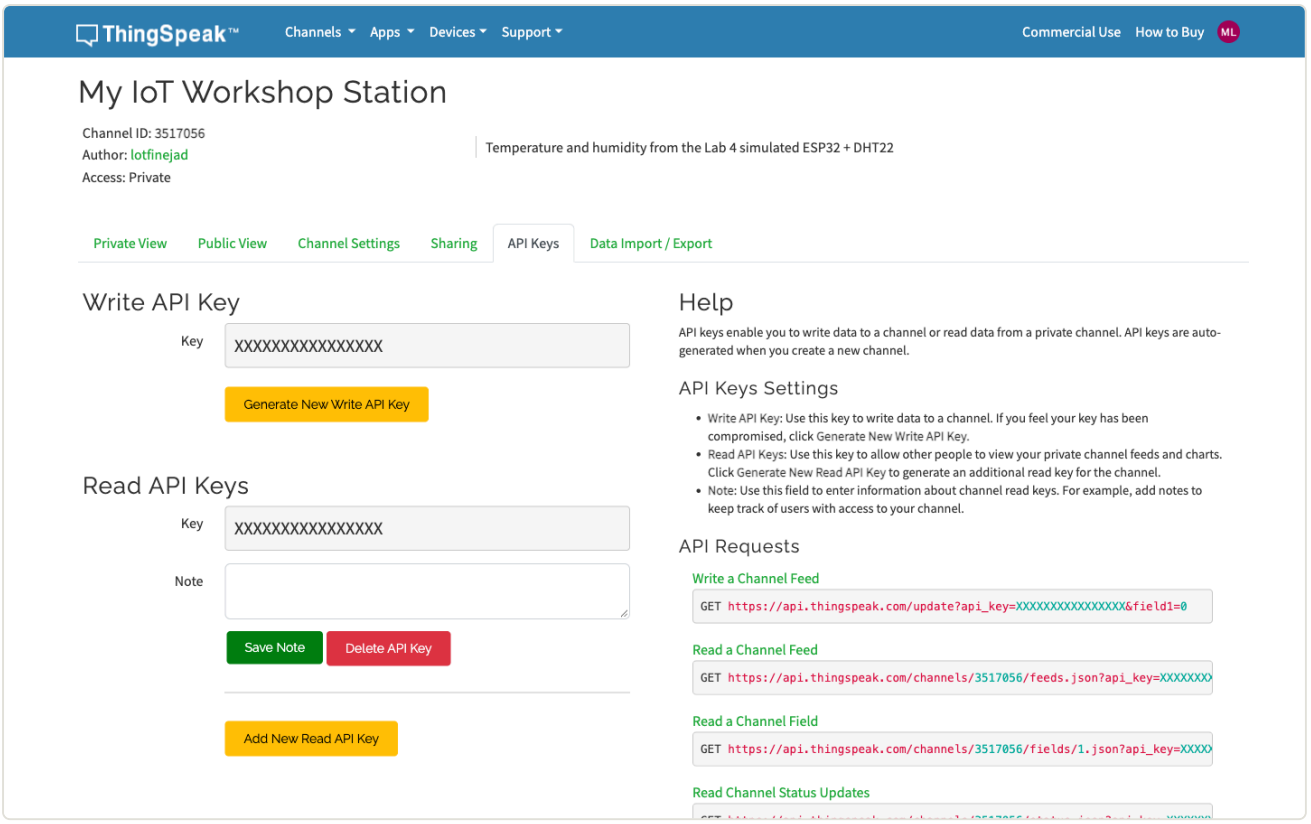
5. Scroll to the bottom of the form and click **Save Channel**.



6. Your new channel's page opens, with a chart for each field. The charts stay empty until your device sends data in Part B (the screenshot was taken afterwards, so its charts already have points). Click the **API Keys** tab.



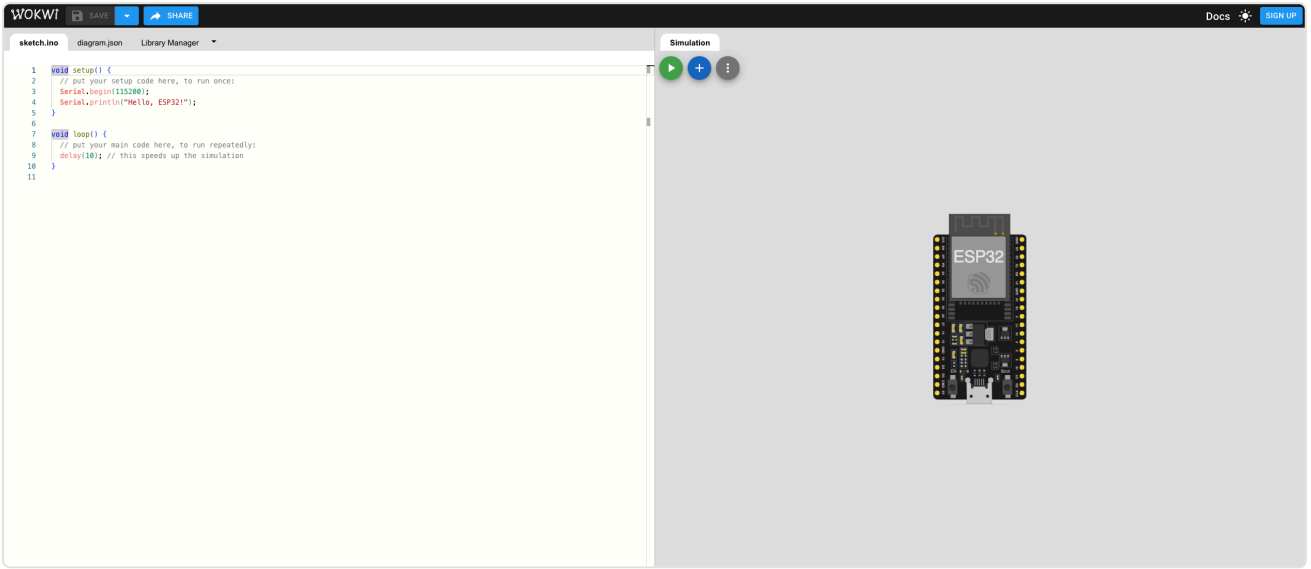
7. Copy the **Write API Key**. You'll need it in the next part.



Treat this key like a password: anyone who has it can write data to your channel. (The keys are hidden in the screenshot for that reason.)

Part B — Send data from the simulator (20 min)

1. Open a new ESP32 project

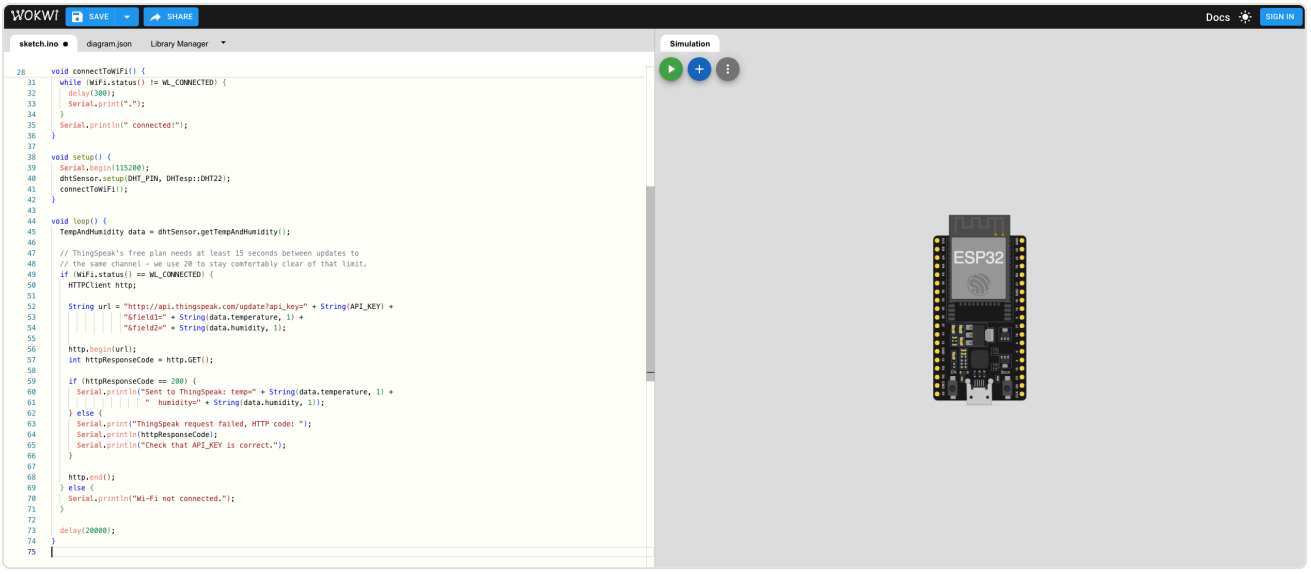


2. Paste the code — with your API key

Copy `sketch.ino` into Wokwi's **sketch.ino** tab. Before pressing Play, find this line near the top:

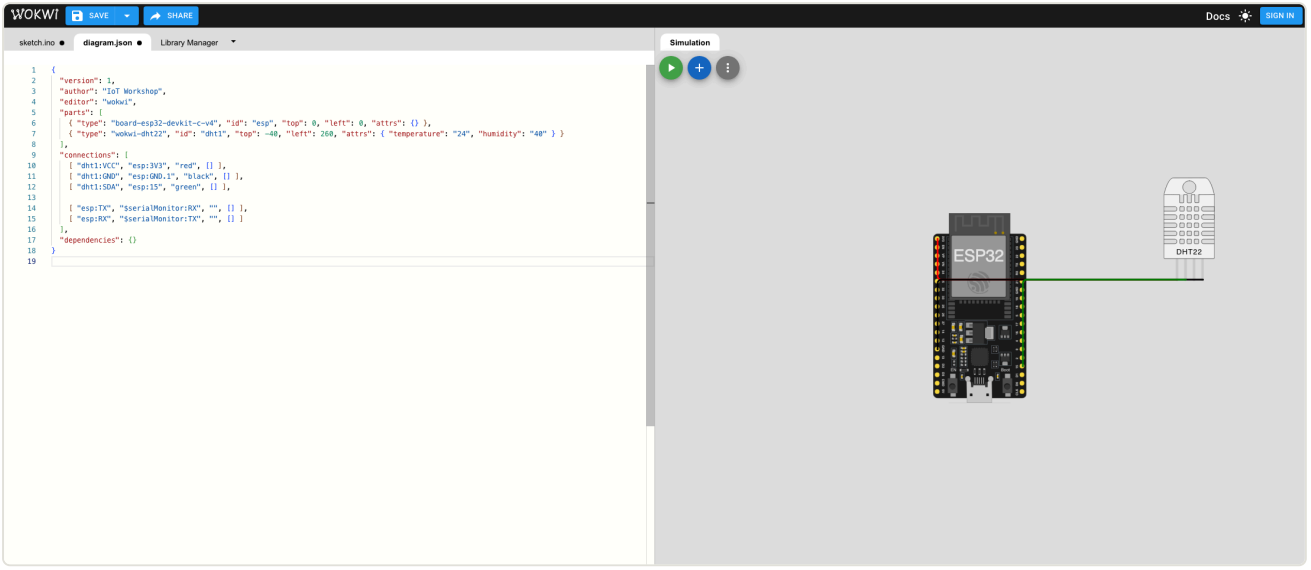
```
const char* API_KEY = "YOUR_WRITE_API_KEY";
```

Replace `YOUR_WRITE_API_KEY` with the key you copied in Part A (keep the quotes).



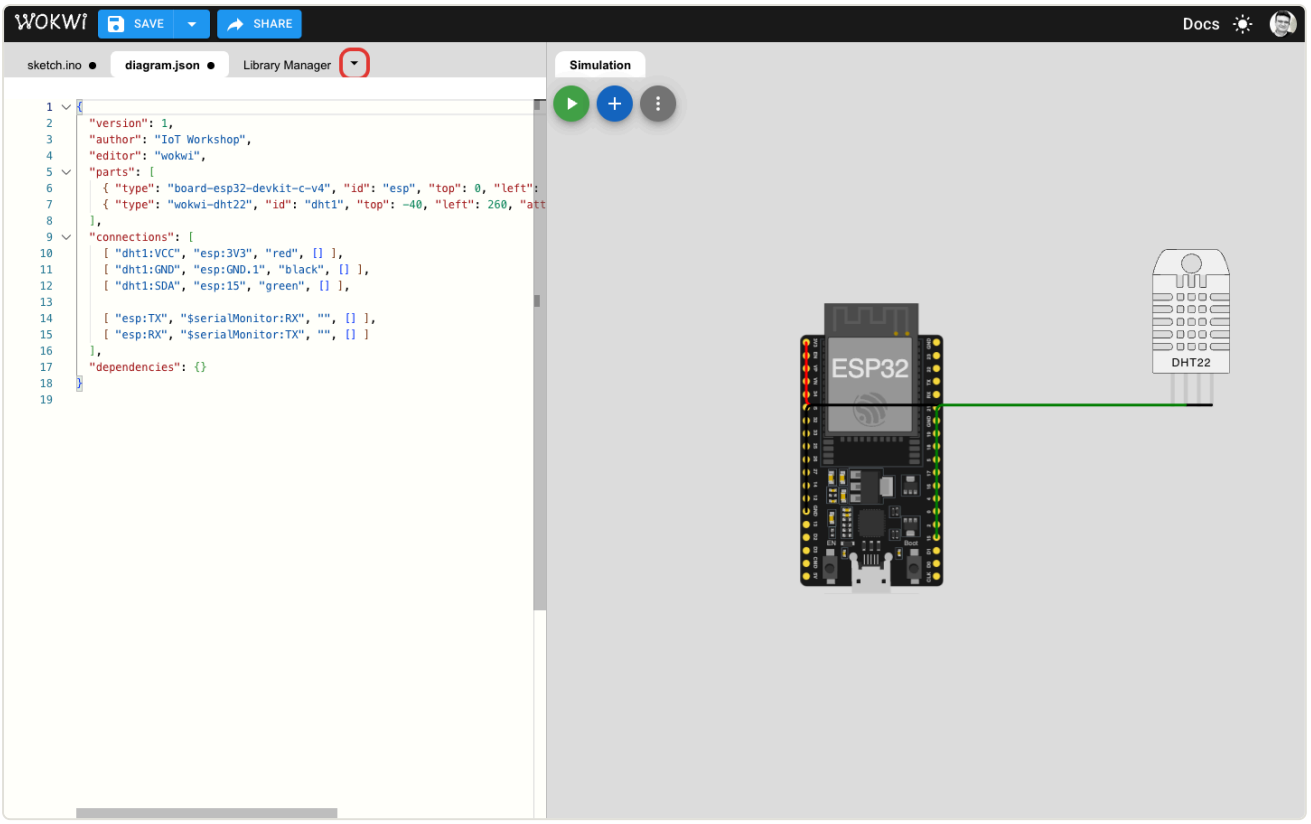
3. Paste the circuit

Copy `diagram.json` into Wokwi's **diagram.json** tab.

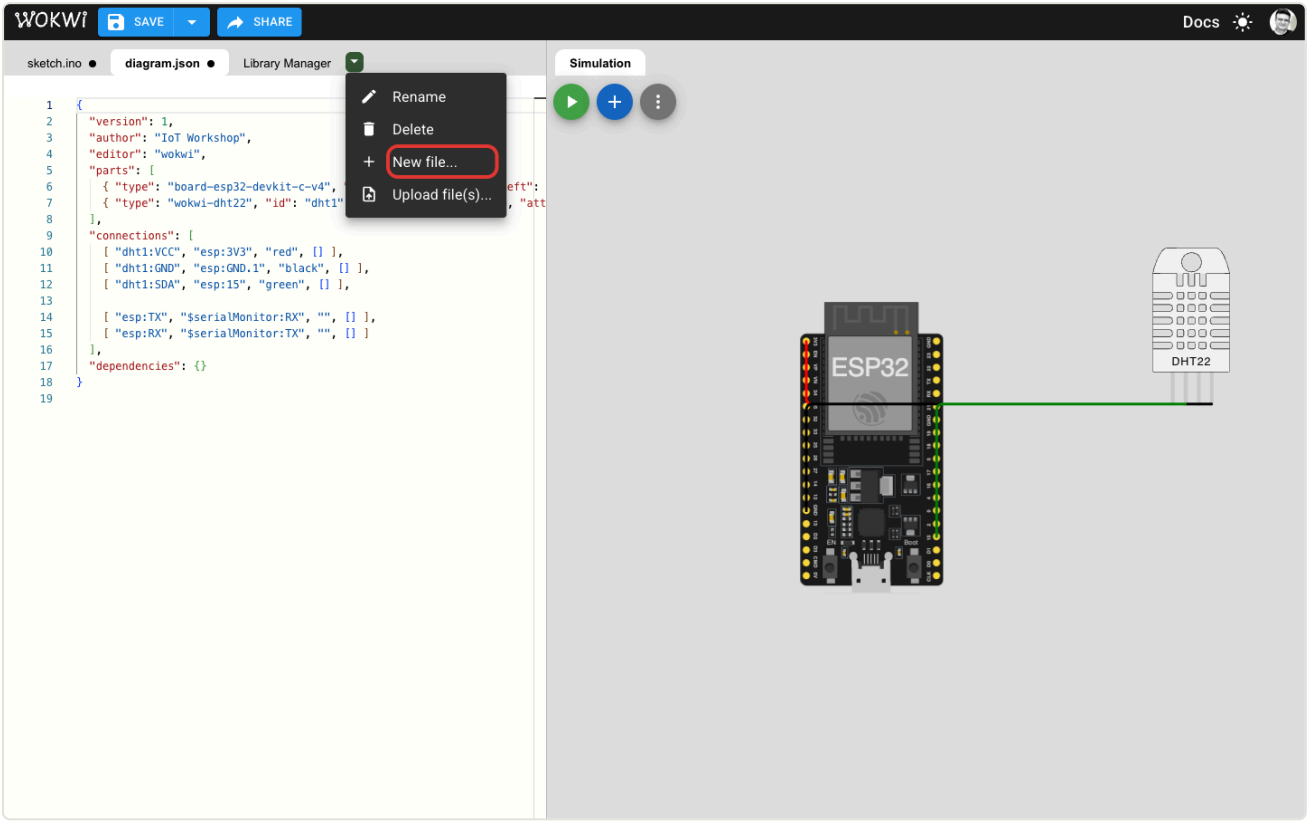


4. Add the library list

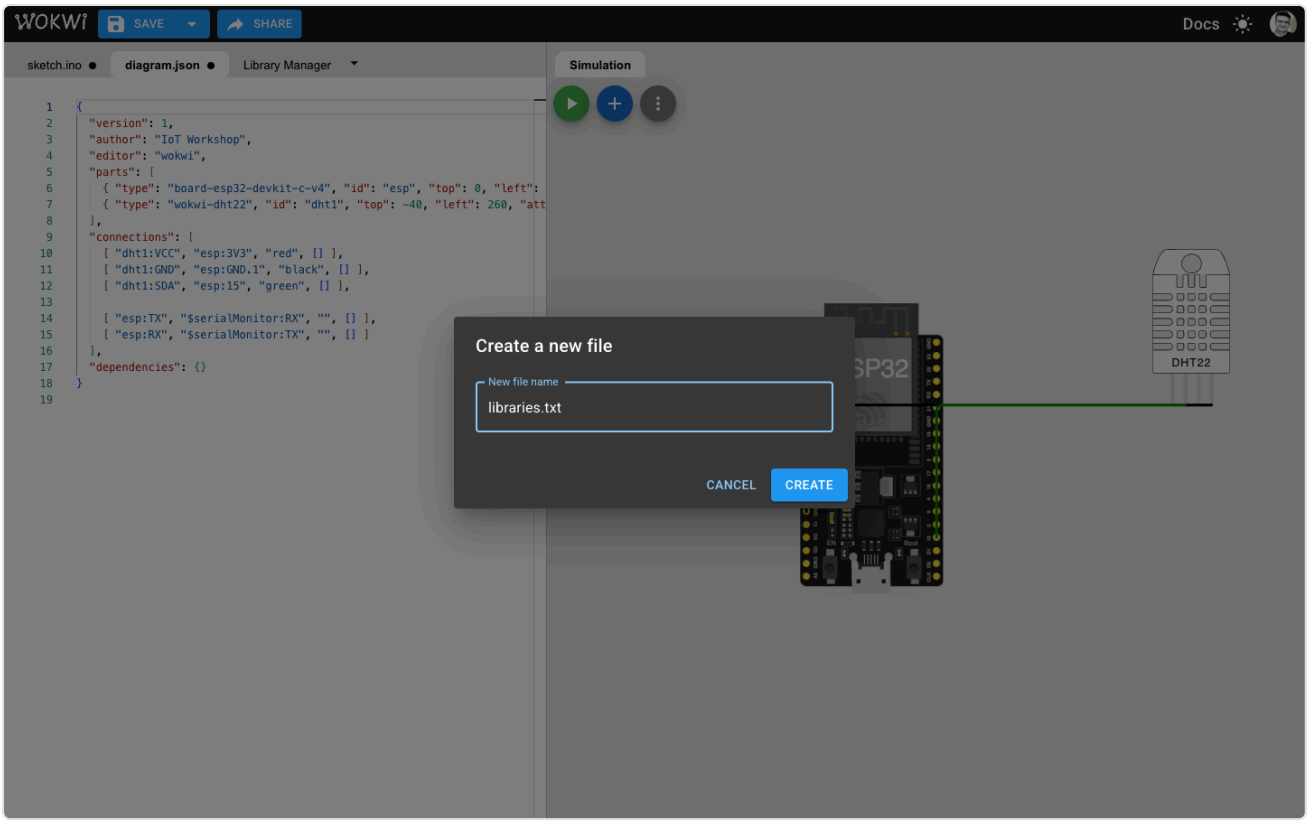
1. Click the small ▾ next to the **Library Manager** tab.



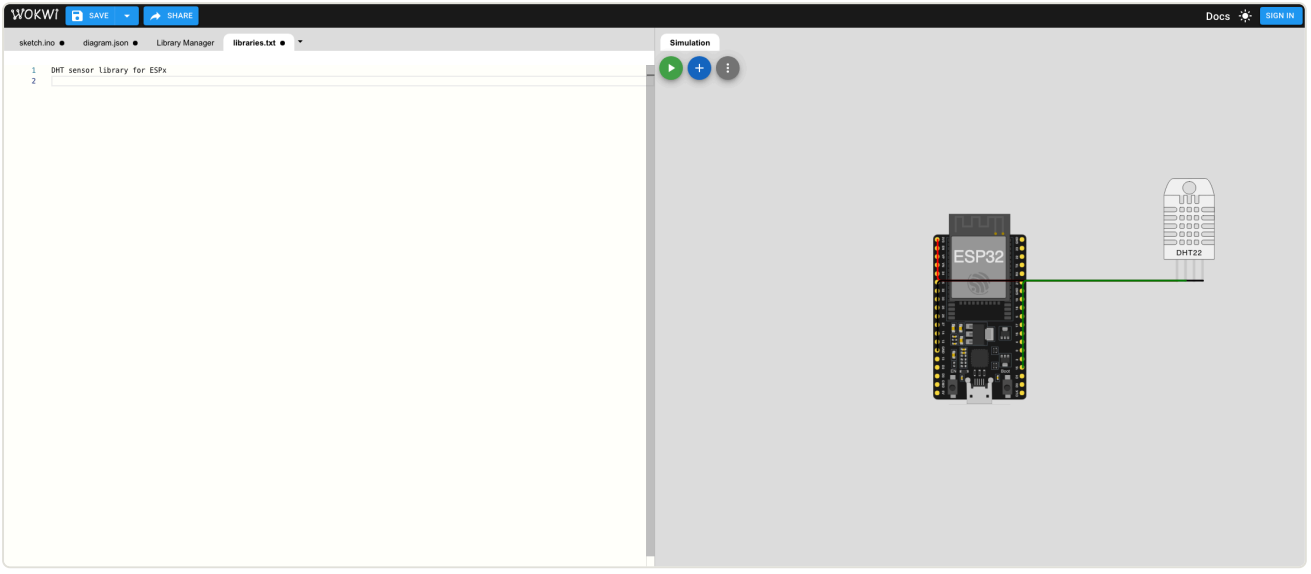
2. Choose **New file...**.



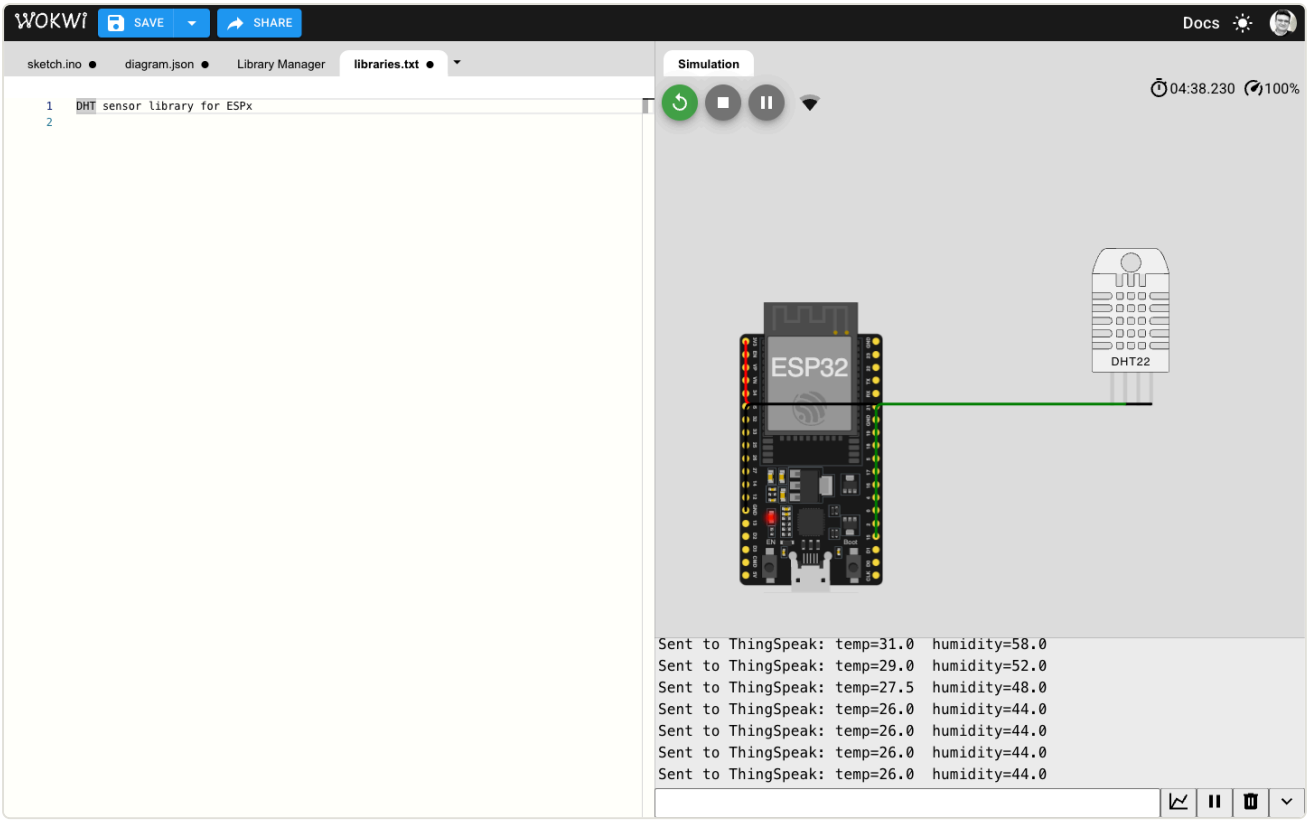
3. Type the name exactly `libraries.txt` and click **CREATE** (or press Enter).



4. Copy the contents of `libraries.txt` from this folder and paste them into the new **libraries.txt** tab.

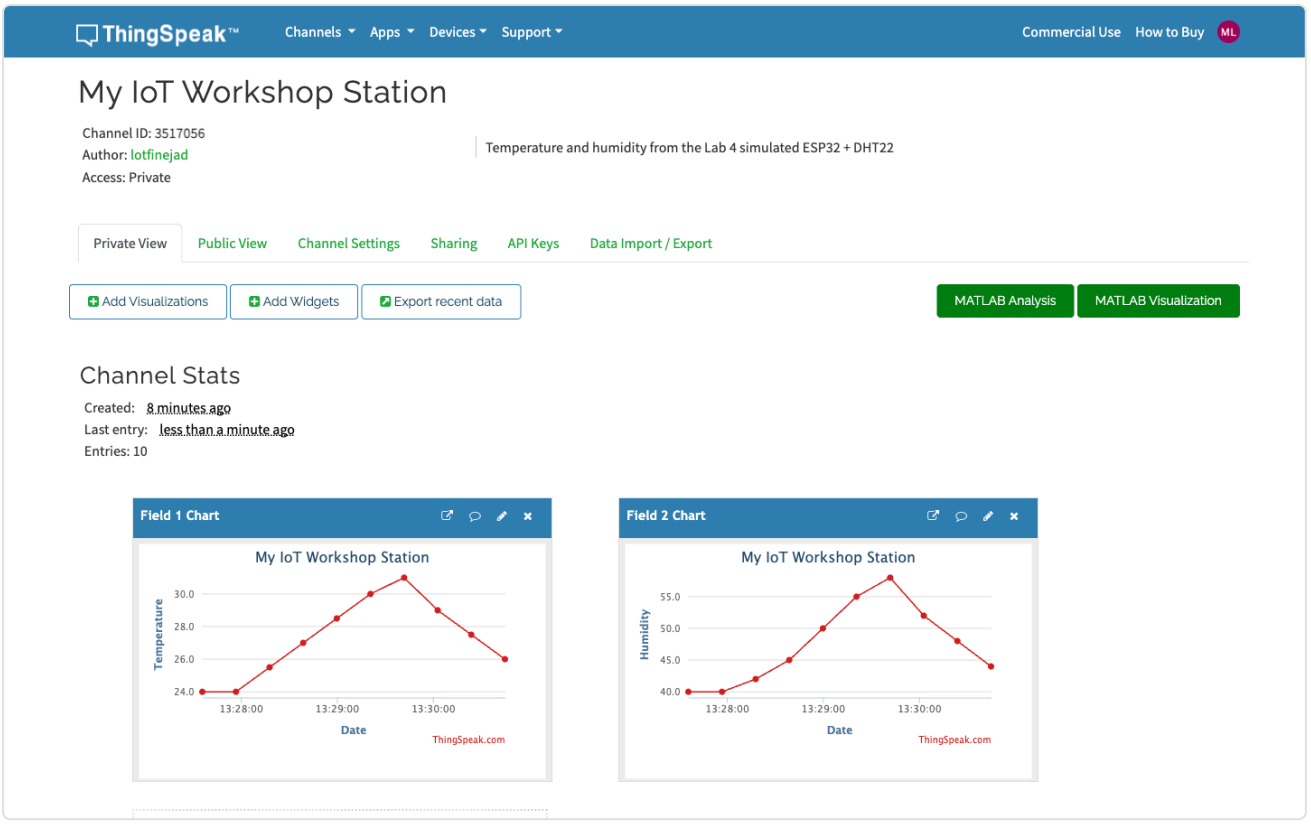


5. Press Play



The Serial Monitor should print `connected!` for Wi-Fi, then start printing `Sent to ThingSpeak: ...` every 20 seconds — or a clear error if the API key is wrong (see Troubleshooting).

6. Go back to your ThingSpeak channel page (**Channels → My Channels →** your channel) and watch **Field 1 Chart** and **Field 2 Chart** — new points should appear as your simulated device sends data. You may need to click the small refresh icon on each chart.



In this example, the DHT22's temperature and humidity sliders were moved between updates (see **Try it** below), so the charts rise and fall. If you don't touch the sliders, you'll see a flat line at 24 °C and 40% — that's correct too.

What you should see

With a correct API key:

```
Connecting to Wi-Fi...... connected!
Sent to ThingSpeak: temp=24.0  humidity=40.0
Sent to ThingSpeak: temp=24.0  humidity=40.0
```

With the placeholder key still in place (i.e. if you skip step 2), you'll correctly see an error instead — that's the code's error-handling working as designed, not a bug:

```
Connecting to Wi-Fi...... connected!
ThingSpeak request failed, HTTP code: 400
Check that API_KEY is correct.
```

Try it

- Click the DHT22 sensor in the Wokwi diagram and drag the temperature slider up and down. Wait 20 seconds and watch your ThingSpeak graph follow.

Understand the code

- `HTTPClient` is the ESP32's built-in tool for making web requests, the same kind your browser makes when you visit a page.
- ThingSpeak's API is deliberately simple: one web address (URL) with the values built right into it — `.../update?api_key=...&field1=...&field2=...`. This style of API, where you fetch a URL to do an action, is what Module 4 called **REST**.
- `http.GET()` returns a numeric response code. `200` means success — this is the same status code your browser gets for a page that loaded fine. If you see a different number, the most common cause is a wrong API key.
- The 20-second delay isn't arbitrary: ThingSpeak's free plan rejects updates sent less than 15 seconds apart on the same channel.

Troubleshooting

- **`Build failed! ... DHTesp.h: No such file or directory`.** You skipped step 4, or `libraries.txt` doesn't contain exactly `DHT sensor library for ESPx`.
- **`ThingSpeak request failed, HTTP code: 400` (or similar).** Your `API_KEY` is missing, wrong, or still the placeholder — go back to Part A step 7 and copy the **Write** key (not the Read key) exactly.
- **Chart doesn't update.** ThingSpeak's free plan needs 15+ seconds between updates on the same channel — the code already waits 20, so give it a full cycle, and try the small refresh icon on the chart itself.

Challenge (if you finish early)

- Add a third field to your channel (e.g. "Heat Index" or just a copy of temperature in Fahrenheit) and send it as `field3` in the URL.
- Look at your channel's **API** tab → **Read** — try fetching your own channel's latest data back with a browser, outside of the code.